

RETAIL LAKEHOUSE MODERNIZATION PROJECT

BUSINESS BACKGROUND

A retail company operates through two independent sales channels.

OFFLINE STORES

Customer, product, and order data are maintained in an Azure SQL Database.

ONLINE E-COMMERCE PLATFORM

Daily order files arrive in Azure Data Lake Storage (ADLS) as JSON files.

Currently, both systems operate independently, resulting in duplicate customers, inconsistent product information, and unreliable reporting.

The company wants to build a modern Lakehouse solution that creates a unified view of customers, products, and sales across both channels.

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SOURCE SYSTEM 1: OFFLINE STORE DATABASE

The offline sales system stores data in Azure SQL Database.

CUSTOMERS TABLE

customer_id	customer_name	email	phone_number	city
101	Ankit Bansal	ANKIT@GMAIL.COM	9876543210	Bangalore
102	Rahul Sharma	rahul@gmail.com	9988776655	Mumbai
103	Priya Mehta	PRIYA@GMAIL.COM	9123456789	Delhi

PRODUCTS TABLE

product_id	sku_id	product_name	category	brand	price
1	SKU1001	Laptop Pro	Electronics	TechMax	85000
2	SKU1002	Wireless Mouse	Electronics	ClickTech	1200
3	SKU1003	Office Chair	Furniture	ComfortSeat	7000

ORDERS TABLE

order_id	customer_id	product_id	quantity	unit_price	payment_mode
OFF10001	101	1	1	85000	Credit Card
OFF10002	101	2	2	1200	UPI

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SOURCE SYSTEM 2: ONLINE E-COMMERCE PLATFORM

Daily order files arrive in ADLS as JSON files.

SAMPLE JSON RECORD

```
{
  "order_id": "ON10001",
  "order_timestamp": "2026-01-10T14:25:10",
  "payment_mode": "Credit Card",
  "customer": {
    "customer_id": "C1001",
    "name": " Ankit Bansal ",
    "email": "[ankit@gmail.com](mailto:ankit@gmail.com)",
    "phone": "+91-9876543210"
  },
  "items": [
    {
      "sku_id": "SKU1001",
      "product_name": "laptop pro",
      "category": "electronics",
      "brand": "TechMax",
      "quantity": 1,
      "amount": 85000
    },
    {
      "sku_id": "SKU1002",
      "product_name": "Wireless Mouse",
      "category": "Electronics",
      "brand": "CLICKTECH",
      "quantity": 2,
      "amount": 1200
    }
  ]
}
```

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BUSINESS CHALLENGES

CUSTOMER UNIFICATION

The same customer may exist in both systems.

Offline System

customer_id: 101

customer_name: Ankit Bansal

phone_number: 9876543210

Online System

customer_id: C1001

customer_name: Ankit Bansal

phone_number: +91-9876543210

Although the customer IDs differ, both records represent the same customer.

Identify same customers based on mobile number(last 10 digits) and unify such customers using a business key.

Customer Attribute Prioritization Rule

When the same customer exists in both systems:

- Match customers using the standardized mobile number.
- If an attribute exists in one system and is missing in the other, use the available value.
- If an attribute exists in both systems, use the value from the Online system as the preferred source.

PRODUCT UNIFICATION

The same product may appear differently across channels.

Offline System

sku_id: SKU1001

product_name: Laptop Pro

category: Electronics

Online System

sku_id: SKU1001

product_name: laptop pro

category: electronics

Students must standardize and unify product information.

Identify same products based on sku_id and unify products information.

Product Attribute Prioritization Rule

When the same product exists in both systems:

- Match customers using the sky id.
- If an attribute exists in one system and is missing in the other, use the available value.
- If an attribute exists in both systems, use the value from the Online system as the preferred source.

ORDER CONSOLIDATION

Orders from both channels must be integrated into a single analytics-ready sales model.

The final solution should support reporting across all sales channels.

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DATA QUALITY CHALLENGES

Examples include:

- Invalid phone numbers (Should be at least 10 digits)
- Missing customer information (email, name etc)
- Missing product identifiers
- Duplicate customer records
- Duplicate product records
- Inconsistent text formatting
- Null business columns
- Customer records with extra spaces
- Email IDs with inconsistent casing
- Product names with inconsistent casing

Invalid records should not be silently discarded.

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EXPECTED Tables (Silver Layer)

CUSTOMER DIMENSION (SCD1)

A unified customer view across online and offline channels.

PRODUCT DIMENSION (SCD2)

A consolidated product dimension with historical tracking of product attribute changes.

SALES FACT TABLE

A combined sales table containing transactions from both channels.

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REPORTING REQUIREMENTS (GOLD Layer)

Business users should be able to answer questions such as:

- What is total revenue by channel?
- Which products generate the highest revenue?
- Who are the highest-value customers?
- What percentage of customers purchase through both channels?
- What are the top-performing product categories?
- How do online and offline sales compare over time?

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INCREMENTAL PROCESSING REQUIREMENT

New online JSON files arrive daily in ADLS.

Offline SQL data is continuously updated in SQL server.

The solution should process only new or changed data and avoid full reloads wherever possible.

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DELIVERABLES

Build a Lakehouse solution that:

- Ingests data from Azure SQL Database and ADLS
- Extracts customer, product, and order information from sql server and nested JSON files (**bronze Layer**)
- Standardizes and cleans data across systems
- Unifies customers and products across channels
- Creates analytics-ready dimensional models
- Supports incremental data processing
- Enables cross-channel sales reporting
- **Lakeflow job(s)** to run the pipeline in a batch mode